Summary of Data

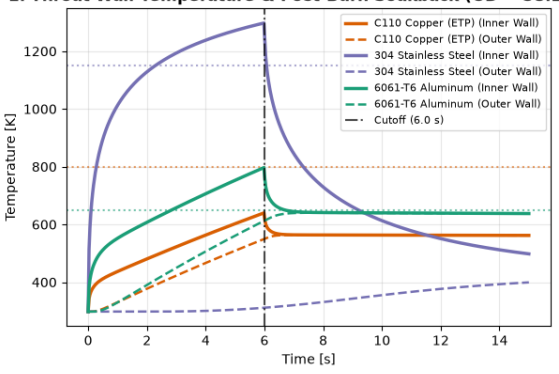
The data shows the diameter sizing using the governing equation for orifice sizing for compressible flow and incompressible flow, in the top table. In the second table it shows the safe firing duration for each common material, using Bartz Equation and Newton's Law of Cooling, (we are aiming for a safe firing duration of 6 seconds). Using the data collected, it was found that copper would be the most thermal conductive thus the best for a safe firing duration while the shell or the outer walls would be best as stainless steel because of its lack of thermal conductivity. Shown in the graphs in relation to time and the temperature showing the inner walls would be much safer made of copper and the main point of the other graphs, except the last graph. Showing the optimized diameter of injector orifices to the percentage of pressure drop for the best amount of holes for both GOX(Oxygen, used as oxidizer) and C3H8(Propane, used as fuel). We are trying to avoid liquid propane as it can clog orifices.

```
================================================================================
INJECTOR ORIFICE SIZING SUMMARY (Manifold = 110.0 PSIA | dP = 20.0 PSID [22.2% Pc] | Cd = 0.65)
================================================================================
Gaseous O2 (GOX)   | mdot = 6.00 g/s | Unchoked       | A_tot =  6.26 mm^2 | d(N=1): 2.823 mm | d(N=2): 1.996 mm | d(N=4): 1.412 mm
Gaseous C3H8       | mdot = 4.00 g/s | Unchoked       | A_tot =  3.65 mm^2 | d(N=1): 2.155 mm | d(N=2): 1.524 mm | d(N=4): 1.078 mm
Liquid C3H8 (SPI)  | mdot = 4.00 g/s | Incompressible | A_tot =  0.53 mm^2 | d(N=1): 0.820 mm | d(N=2): 0.580 mm | d(N=4): 0.410 mm
================================================================================

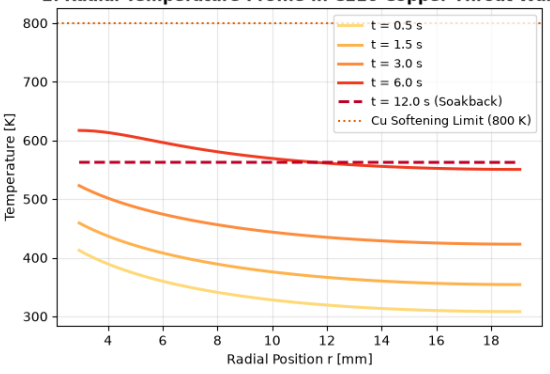
================================================================================
TRANSIENT HEAT-SINK SAFE FIRING DURATION (Throat ID = 5.89 mm, Body OD = 38.1 mm, Test Burn = 6.0 s)
================================================================================
C110 Copper (ETP)      | Max Inner T =  639.9 K | Soakback Outer T = 561.9 K | Limit (800 K) Reached At: > 6.0 s (Safe)
304 Stainless Steel    | Max Inner T = 1297.4 K | Soakback Outer T = 399.4 K | Limit (1150 K) Reached At: 2.28 s
6061-T6 Aluminum       | Max Inner T =  796.9 K | Soakback Outer T = 637.2 K | Limit (650 K) Reached At: 2.86 s
================================================================================
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GOX/C3H8 Torch Igniter: 1D Transient Heat-Sink & Injector Orifice Design Suite

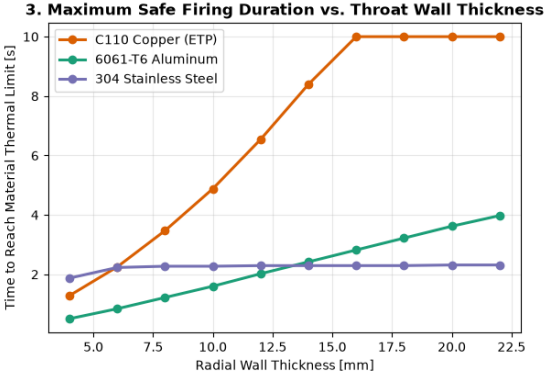
1. Throat Wall Temperature & Post-Burn Soakback (OD = 38.1 mm)



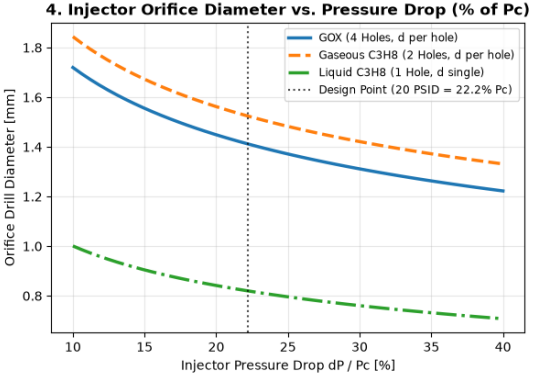
2. Radial Temperature Profile in C110 Copper Throat Wall



3. Maximum Safe Firing Duration vs. Throat Wall Thickness



4. Injector Orifice Diameter vs. Pressure Drop (% of Pc)



Summary of Data

The tables at the top display the chamber geometry of the ignitor, this was found using Fournier's Law for Cylindrical Coordinates, and a CEARUN simulator to find the best data points for our design. In graph 1, it shows what should happen to the gas as it enters the throat at mach 1, and as it increases through the exit. Graph 2 shows how the static pressure inside of the chamber changes as the temperature decreases. In graph 3, the velocity of the exiting gas or mixture of C3H8 and O2 increases as it gets to the throat and outwards to show a direct relationship between graph 1 and graph 3. Lastly in graph 4, it shows the shifting on the molecular level of how as the C3H8 and O2 approach the throat the reaction begins happening rapidly change the mass fractions and showing what is being exhausted from the chamber/ignitor.

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================================================================================
OPTIMIZED TORCH IGNITER GEOMETRY & SPREADSHEET AUDIT
================================================================================
Mass Flow Split      : Total = 10.00 g/s | GOX = 6.00 g/s | C3H8 = 4.00 g/s
Mixture & Stoich     : O/F = 1.50 (40% Fuel) | Stoich O/F = 3.6281 | PHI = 2.4188
--------------------------------------------------------------------------------
Throat Area (At)     : 2.72493e-05 m^2 (27.2493 mm^2)
Throat Diameter (dt) : 5.89024 mm
Chamber Diameter (dc): 15.0172 mm  [FIXED: dt * sqrt(6.5) instead of dt * 6.5 = 38.287 mm]
Chamber Area (Ac)    : 177.1204 mm^2 (Contraction Ratio Ac/At = 6.50)
Exit Diameter (de)   : 7.3699 mm  (Exit Area Ae = 42.6588 mm^2, Ae/At = 1.5655)
--------------------------------------------------------------------------------
Vc (Spreadsheet Err) : 117.8047 mm^3 [Caused by multiplying dt * L*]
Vc (True @ L*=20 mm) : 544.9857 mm^3 (Residence time = 0.027 ms -> Too short!)
Vc (Optimized Design): 4155.52 mm^3 (L_cyl = 20.0 mm, Effective L* = 152.5 mm)
Residence Time (Opt) : 0.207 ms  (7.8x longer residence time for stable spark kernel)
Igniter Performance  : Thrust = 19.29 N | Total Plume Thermal Output = 50.2 kW
================================================================================
```

